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Study of biotic interaction of stem parasite : Cuscuta sp.

A part from microbial pathogens and herbivoreous autotraphs plant are also parasized by other plants. Parasitic plants keep their host alive until their life cycle is completed, and keep on feeding from their host. Most of the parasitic plants life cycle completes earlier than the one of the host plant which leads to a premature death of the parasitized plants and thus can cause crop damage.

Like other angiosperms, the life cycle of cuscuta sp begains with seed germination. Germinating budlings unable to survive along for a longer time and must find a parasite host plant stem within the first few days. To find a potential hosts, Cuscuta recognised plant varieties with volatile as chemo-attracti-ons which guid seedlings growth and increase the chances of successful infections.

After finding an appropriate host plant, the first physical contact initiates an attachment phase, in which the parasitic epidermal and parenchymal cells begin to differentiate into a secondary meristem and develop prehaustria, also known as adhesive disk. Important signal initiating and compraling the pre haustria formation including mechanical pressure, osmotic pressure and phyto-hormones such as cytokinin and auxin. The pre haustrical cells starts to produce and secreate adhesive substance such as pectins and other polysaccharides, reinforcing the addesion, within the

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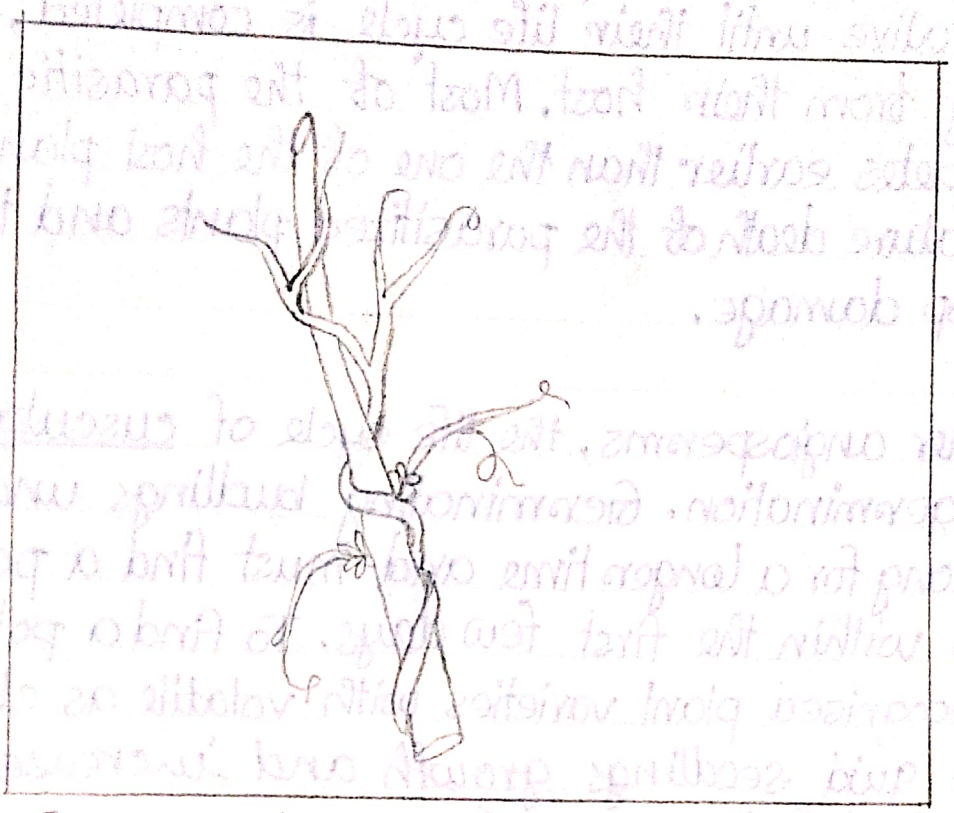


Fig: Cuscuta sp. on a host plant

first hour of contact, cuscuta also includes the host plant to produce its own sticky substance such as pectins and poly-saccharides, reinforeing the adhesion. cells of the invading haustoria form searching hyphae, which try to reach phloem or xylem cells of the host plant's vascular bundles. To form a connection to the xylem, parasitic and host plant cells of the xylem parenchyma commence a synchronised develop-ment, fusing to built a continious xylem tube form the host to a parasitic, with functional connectional to the xylem and phloem of its host. The parasitic plant is supplied with water nutrients and carbohydrates.

Morphology :-

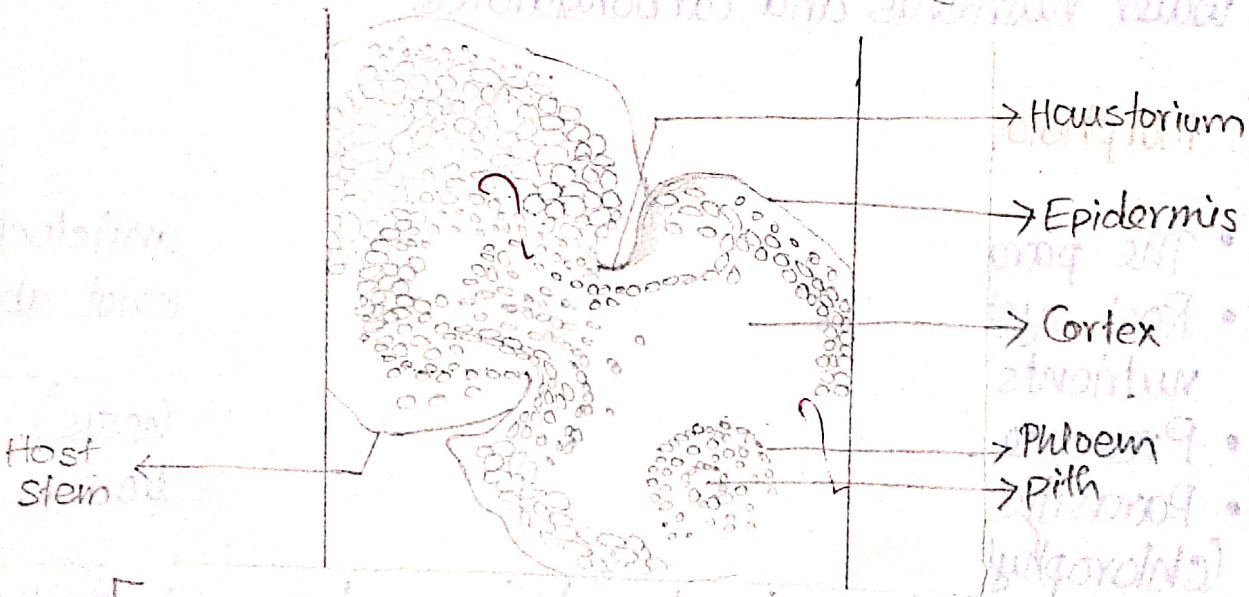
- The parasitic plant entwines the host plant anticlockwise.
- Root system is missing entirely (as they donot absorb nutrients from soil.
- Presence of scale leaves (reduces photosynthesis).
- Parasitic stem colour ranges cream to yellow. (Chlorophyll is unnecessary).
- The haustorium developed from stems, is embedded inside the host, serving to feed the parasitic. located externally to the host.

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[Fig: T.S of host along with Cuscuta sp] (microscopic photoplate view)



[Fig- Haustorium entering host plant]
(microscopic view labelled diagram)

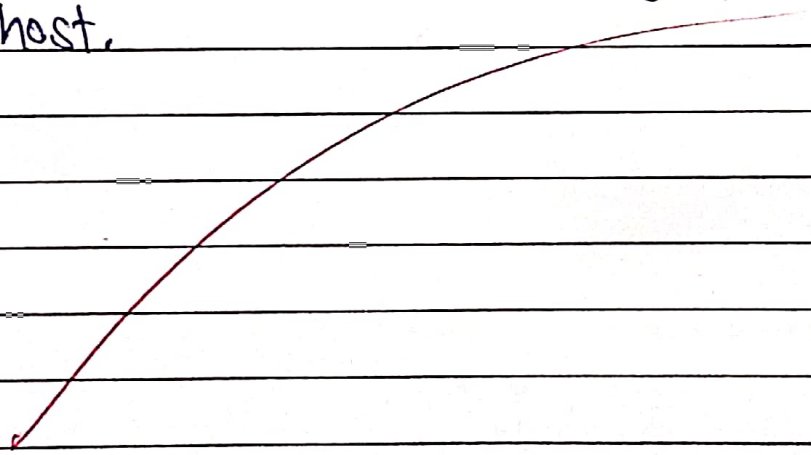
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Anatomy –

A transverse section through both plants at the place where the dodder stem is attached to its host shows that the parasite stem is fixed to the host stem by a disk the superficial cells of which are gratdy entered and form its disk, a housterium has grown into host stem tissue. Some of the housteria are seened applied to the vascular bundle of the host, each housterium has a central xylem ~~stem~~ stand with spherical vessels, continuous with the bundle of the ~~dogger~~ dodder stem.

In favourable section this strand may be traced right into the xylem of the host plant, while the housteria have also elements which join on to the phloem of the host.



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